Monte Carlo Simulation of Spurious Regression

This project investigates how reliable hypothesis testing is when regressing two independent time series. Two AR (1) processes were simulated: one for y_t and one for x_t ,where each depends on its past value and a random error term. The error terms were generated independently at all leads and lags, ensuring that the two series are unrelated. As a result, the true value of the regression coefficient is zero, meaning there should be no relationship between the two variables.

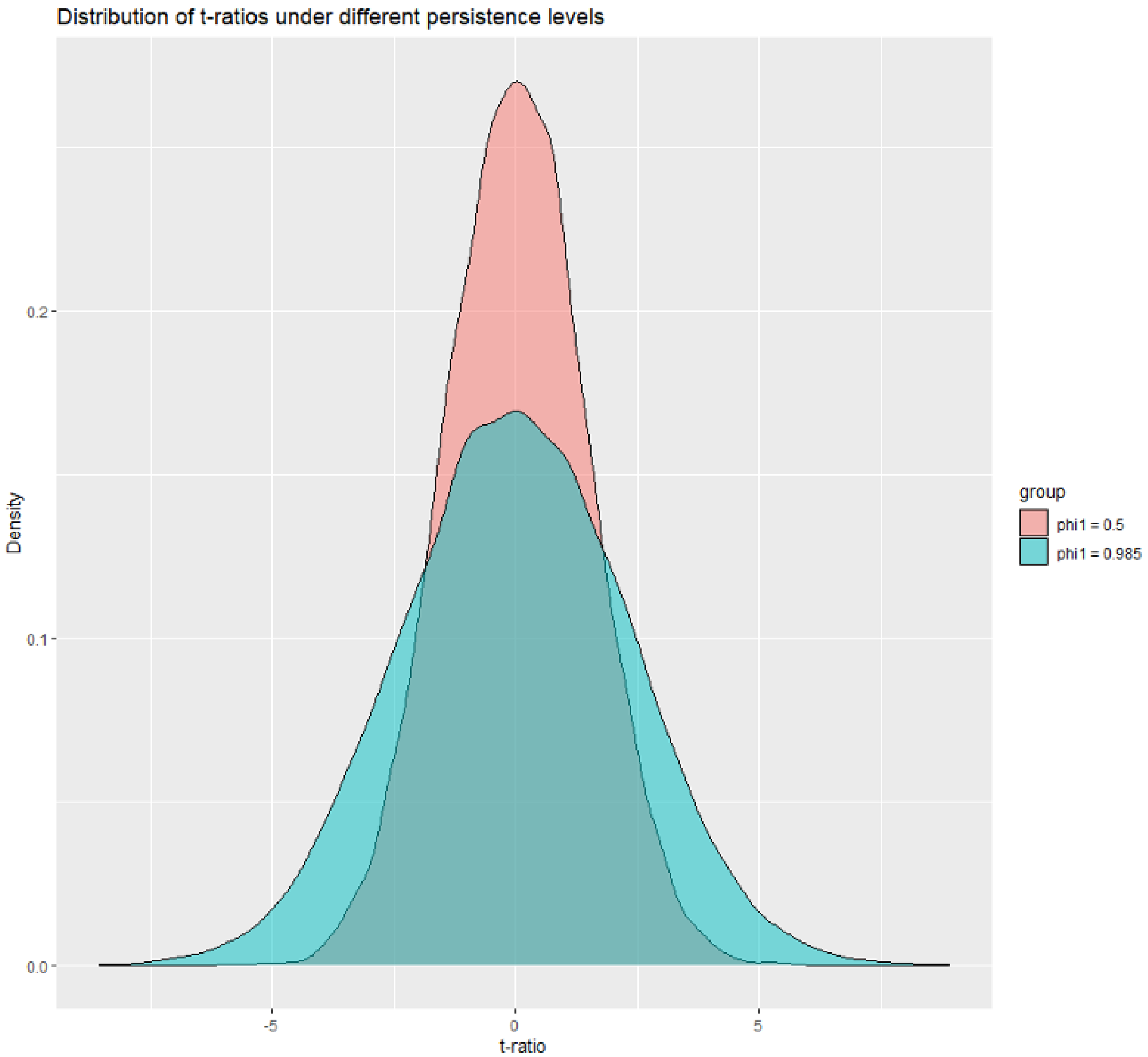
The regression model estimated is $y_t = \alpha + \beta x_t + u_t$. The coefficients were calculated using the ordinary least squares (OLS) formula $\hat{\beta} = (X'X)^{-1}X'y$. From this, the standard error and corresponding t-statistic were computed to test the null hypothesis $H_0 : \beta = 0$. The persistence of the x_t regressor was fixed at **ϕ**2 =0.7, while the persistence of y_t, **ϕ**1, was varied across a range of values from 0.5 to 0.985. This allowed the effect of increasing persistence on the behaviour of the t-statistic to be examined.

The simulation was implemented using nested loops. The outer loop iterates over the different values of **ϕ**1, while the inner loop repeats the simulation r=15,000 times for each value. In each iteration of the inner loop, new time series for x_t and y_t are generated, the regression is estimated, and the coefficient estimate, t-statistic, and rejection decision are stored. This process builds up the sampling distribution of the estimator and the t-statistic. To ensure efficiency, a smart programming approach was used by pre-allocating storage matrices for these values before entering the inner loop. This avoids repeatedly resizing objects during execution, which can slow down the code. In addition, values that do not change, such as the number of regressors, were defined outside the loop to avoid unnecessary repetition.

The results show that the mean of the estimated coefficient remains close to zero for all values of **ϕ**1, which confirms that the OLS estimator is unbiased. However, the variance of the estimates increases as **ϕ**1 becomes larger. This means that the estimates become more spread out when the series is more persistent. More importantly, the proportion of times the null hypothesis is rejected at the 5% significance level increases as persistence increases. For lower values of **ϕ**1, the rejection rate is close to the expected 5%, which is what we would expect when the null is true. However, for higher values of **ϕ**1, the rejection rate rises well above 5%, showing that the test is rejecting too often.

This pattern is clearly visible in the density plots of the t-statistics. As shown in **Figure 1**, when **ϕ**1=0.5, the distribution is tightly centred around zero and has a shape similar to a normal distribution. Most values are close to zero, and only a small number fall beyond the critical values. In contrast, when $\phi 1=0.985$, the distribution is much wider and more spread out, with heavier tails. This means there are more extreme values, and the t-statistic is more likely to exceed the critical value of ±1.96. As a result, the null hypothesis is rejected more often, even though it is true.

Figure 1:



Overall, these results demonstrate the problem of spurious regression. Even though the two-time series are independent, high persistence makes them appear related when standard regression methods are used. The OLS estimator itself remains unbiased, but the associated hypothesis test becomes unreliable. In particular, the increase in variance leads to more extreme t-statistics, which causes the null hypothesis to be rejected too frequently. This shows that statistical significance can be misleading when working with highly persistent data.

This has important implications for real-world applications, especially in economics and finance, where time series data often show elevated levels of persistence. If this issue is not considered, it can lead to incorrect conclusions about relationships between variables. Therefore, care must be taken when interpreting results from regressions involving persistent time series.